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A physically-informed hybrid deep learning framework for multi-timescale voltage prediction of proton exchange membrane fuel cells under dynamic loading

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HIGHLIGHTS

- A physically-informed hybrid deep learning framework is proposed for PEMFC voltage prediction.
- Lévy jump-enhanced two-layer LSTM captures nonlinear transient voltage fluctuations.
- TCN–LSTM achieves robust long-term degradation prediction with lower computational complexity.
- Multi-timescale prediction is validated using experimental and IEEE 2014 PEMFC datasets.
- Hybrid fusion of physical, signal-processed and deep features improves prognostic reliability.

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ABSTRACT

Proton Exchange Membrane Fuel Cells(PEMFCs) are widely adopted in clean energy systems; however, accurate short-term and long-term voltage prediction is required for reliable operation under dynamic loading conditions. This work proposes a physically-informed hybrid deep learning framework for multi-timescale voltage prognostics. Short-term transient voltage prediction is validated using experimentally acquired data from a 100 W PEMFC test bench, while long-term degradation prediction is evaluated using the IEEE 2014 benchmark dataset. The approach combines thermofluid features with advanced deep learning architectures for accurate voltage prediction in PEMFC systems. For long-term degradation prediction, multiple models including Auto encoder–Deep Neural Network (AE–DNN), Temporal Convolutional Network–Long Short-Term Memory (TCN–LSTM), Quantum LSTM (QLSTM) and Transformer–LSTM are systematically evaluated. TCN–LSTM framework is adopted to model gradual degradation dynamics while maintaining computational efficiency and stable temporal learning demonstrating effective balance between prediction accuracy and computational efficiency. For short-term transient voltage prediction, a Lévy jump-enhanced and optimized two-layer LSTM is employed to capture abrupt nonlinear voltage fluctuations during step-load variations. Signal pre-processing and Savitzky–Golay filtering preserves dynamic characteristics while reducing noise. The proposed framework offers a scalable solution for real-time voltage prediction, condition monitoring and digital twin applications.

Nomenclature	Description
PEMFC	Proton Exchange Membrane Fuel cell
TCN	Temporal Convolution Network
LSTM	Long Short Term Memory
DNN	Deep Neural Network
QLSTM	Quantum Long Short term Memory
AE	Auto Encoder

1. Introduction

1.1. Background

PEMFCs are becoming an essential part of next-generation clean

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energy applications, including microgrid, stationary power systems and electric cars. They are well-suited for AI-driven energy management systems as well as transportation electrification due to their high energy density, quick reaction time and emission-free operation [1]. Reliable performance and long-term operational stability are crucial as PEMFCs are being incorporated into hybrid and fully electric platforms more frequently as the global movement toward sustainable energy picks up speed. Although PEM fuel cells offer significant potential, their widespread deployment is limited by nonlinear voltage degradation and dynamic transient behaviour [2]. Variations in operating conditions, such as dynamic load changes, temperature fluctuations and reactant variability, accelerate performance degradation, thereby reducing stack efficiency and operational lifetime. Traditional voltage monitoring and control methods often fail to capture long-term degradation trends, particularly under noisy real-world operating conditions. To predict voltage degradation, improve control strategies and make predictive maintenance possible, accurate and strong prognostic models are necessary. Such models improve the overall system reliability and extend the fuel cell life time [3]. Precise modelling for prediction of degradation in terms of the fuel cell voltage and its corresponding state of health (SOH) or health index (HI) is important to ensure proper functioning of the fuel cell system and to implement effective maintenance procedures that can prolong the service life of the system [4–8].

Instead of relying on just one method, the system that combines the physical model, filtering and voltage prediction through a hybrid fusion framework is to be developed. This improves reliability and reduces uncertainty, because each approach compensates for the limitations of the others in prediction of voltage under different environment. Finally, the combined system provides practical outputs such as fault detection, state-of-health (SOH) estimation, condition assessment and degradation monitoring.

Auto encoders (AE) are often used to get hidden features from fuel cell voltage and operational data that are sensitive to degradation. Hybrid architectures like AE-DNN and AE-LSTM first turn high-dimensional measurements into smaller, easier-to-use representations. Then regression or sequential modelling is performed on the model. AE-LSTM frameworks, in particular, use LSTM's ability to model time to track how voltage changes over time in different operating conditions. These methods have shown that they can make better predictions and are less affected by noise than direct LSTM or DNN models. Their latent features are often used to estimate health indices that are related to underlying degradation mechanisms [4] [5]. As an alternative to regression-based forecasting with LSTM The fault detection model was transformed by the authors into a many (each sequence) to one (unique fault) classification issue. Advanced LSTM variants, including Quantum LSTM (QLSTM) and attention-enhanced LSTM, explores the fuel cell degradation forecasting. These models improve the memory and interpretability of LSTM networks by emphasizing relevant temporal features or reducing model complexity. Such models have demonstrated strong performance for both voltage prediction and health index estimation in datasets with highly nonlinear and stochastic degradation behaviour [7] [8].

TCN layers efficiently capture short-term temporal correlations in voltage sequences similar to PCA (Principal Component Analysis), while the LSTM layers model the long-term degradation trends. Recent studies have shown that TCN-LSTM architectures outperform standalone RNNs or CNNs for long-horizon voltage prediction, particularly under noisy and dynamic operating conditions [9]. TCN-LSTM hybrids combine the local feature extraction capabilities of Temporal Convolutional Networks (TCN) with the long-term dependency modelling of LSTMs [9–12].

Transformers, with their self-attention mechanism, can capture long-range dependencies without suffering from the vanishing gradient issues of traditional RNNs. In fuel cell applications, Transformer architectures have been used to model long-term voltage degradation trends. Transformer - LSTM hybrids further combine the global context modelling of Transformers with the sequential memory of LSTMs, enabling robust prediction of voltage and SOH even under irregular and noisy operational patterns [13] [14].

A multitask deep learning framework based on Long Short-Term Memory with Temporal Pattern Attention (Multi-LSTM-TPA) is proposed for dynamic modelling of PEMFC systems. This approach addresses the limitations of traditional physics-based models by leveraging an end-to-end learning strategy capable of capturing nonlinear and coupled electrochemical dynamics. The incorporation of the Temporal Pattern Attention (TPA) mechanism allows dynamic weighting of temporal features, enhancing the model's ability to capture multi-timescale behaviour as reported in Ref. [15].

1.2. Limitations of existing methods

Recent research incorporates probabilistic methods into time-series prediction. Alongside providing probabilistic, risk-informed performance forecasts appropriate for automotive prognostics and health management [4], present a cutting-edge, uncertainty-aware degradation prediction framework for PEM fuel cells that successfully bridges the gap between deterministic data-driven models and real-world driving variability. A Lightweight Transformer based QSLSTM Framework for Real-Time Degradation Index Prediction of PEM Fuel Cells Using Embedded Sensor Data is deployed in Ref. [5]. Voltage prediction is necessary because voltage is the most sensitive and informative indicator of fuel cell performance, health, efficiency, safety, and remaining lifetime. Short-term transient voltage dynamics continue to be essential to embedded applications and control, despite long-term deterioration.

Physics-based analyses reported in Refs. [16], [17–22], which created transient electrochemical and thermodynamic representations, are the origin of dynamic modelling techniques. With a focus on temporal parameter fluctuation [23], investigated voltage behaviour during flooding and drying. Hybrid digital twin concepts integrating hydrogen production and fuel cell systems were introduced in Ref. [14], supporting real-time monitoring applications. Open-source physics-based embedded models such as Alpha PEM [19] provide computationally efficient alternatives for real-time systems.

The computational complexity of the Transformer encoder scales quadratically with sequence length $O(T^2)$ due to the self-attention mechanism [24], whereas LSTM-based models scale linearly with sequence length $O(Th^2)$. Quantum Machine Learning Algorithm allows for the management of large databases and the execution of complex computational tasks more efficiently than classical algorithms. In hidden dimension, quantum-inspired matrix operations usually result in polynomial order increases as reported in Ref. [25]. In contrast to Ref. [15], the present work emphasizes multi-timescale voltage prediction using both experimentally acquired transient data and benchmark degradation datasets, providing a complementary perspective for PEMFC prognostics under dynamic load conditions. Recent research on non-Gaussian priors for stochastic processes shows that non-Gaussian heavy tailed processes and Lévy-type processes give a better description of sudden and non-smooth changes than the Gaussian approach as given in Refs. [26–28].

1.3. Motivation

Voltage prediction in a Proton Exchange Membrane Fuel Cell (PEMFC) system is a critical parameter for ensuring the reliability, efficiency, and durability of the fuel cell. In real-world applications, fuel cells are always exposed to dynamic conditions such as load changes, temperature gradients, reactant depletion and humidity variations, which have a direct impact on the electrochemical properties of the fuel cell. Because stack voltage encapsulates activation, ohmic and mass transport losses, it serves to be the most sensitive and integrative indicator of internal physicochemical states. However, conventional empirical or purely data-driven models often fail to maintain the

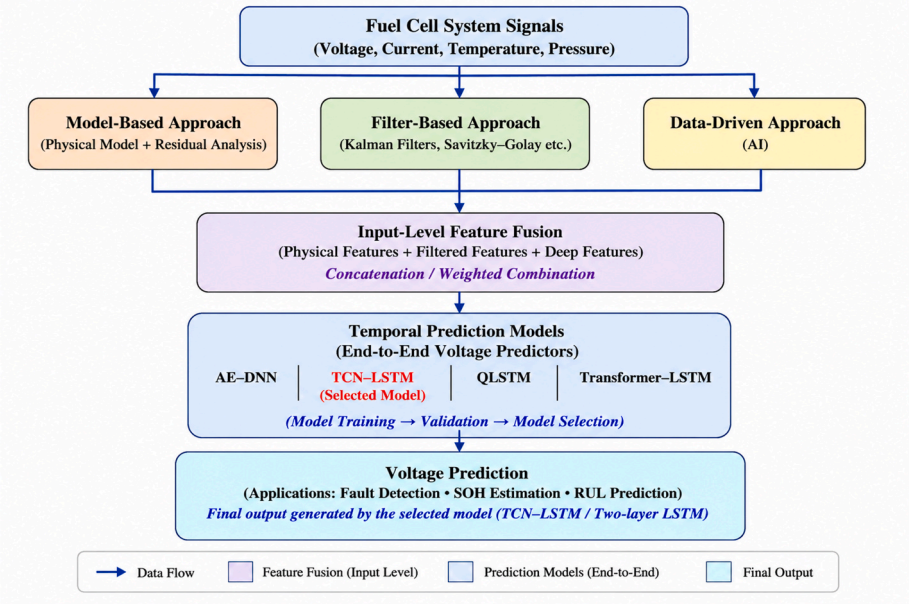


Fig. 1. Proposed physically informed hybrid feature framework for voltage prediction of fuel cell.

robustness when exposed to unpredictable operating regimes, as they neglect the underlying thermo-fluid interactions governing membrane hydration, heat transfer and reactant transport. This limitation motivates the development of a thermo-fluid feature informed voltage prediction framework that embeds physically meaningful operating descriptors into hybrid deep learning architectures. This work suggests fusion of data-driven methods with real time data enhances predictive accuracy and generalization thereby can be adopted for early fault detection, degradation-aware control and digital twin integration [28]. Ultimately, improving voltage prediction reliability is essential for accelerating the commercialization and intelligent management of next-generation PEM fuel cell systems. However, most existing approaches focus either on short-term transient modelling or long-term degradation prediction independently. A unified multi-timescale framework integrating both experimental transient data and long-term degradation datasets remains limited in current literature. Moreover, a mathematically formulated multidimensional fusion method is proposed for the integration of thermofluid features, signal-processed features and deep learning features through cascade and attention-based approaches.

The main contributions of this work are:

1. Physically-informed Multi-timescale PEMFC voltage prediction using experimental transient data and IEEE 2014 degradation dataset.
2. Lévy jump-enhanced LSTM for modelling non-Gaussian voltage transients under dynamic load variations.
3. Benchmark evaluation of hybrid deep learning architectures for robust multi timescale PEMFC voltage prediction under dynamic conditions.

Section 2 describes the proposed hybrid deep learning framework, detailing data pre-processing, feature representation, model configurations, and the unified multi-timescale approach for short-term and long-term voltage prediction. Section 3 deals with results and discussion. Section 4 deals with conclusion.

2. Methodology

2.1. Hybrid deep learning architectures for voltage prediction in long and short term voltage prediction

The developed framework implements the methodology of multi-scale model, where the task of short-term transient forecasting and long-term degradation forecasting is seen as two parts that form a whole of voltage forecasting task. The short-term part addresses the issue of fast voltage dynamics that occur at the time of load switching; on the other hand, the long term part covers the slow process of voltage degradation. The proposed framework integrates signal processing and deep learning components in a sequential pipeline. Raw sensor data are first pre-processed using filtering and normalization techniques to remove noise and stabilize input distributions.

The processed signals are then fed into deep learning architectures for feature extraction and temporal modelling. In the proposed multi-timescale framework, two layered–LSTM is adopted for short-term transient voltage prediction because recurrent memory mechanisms are capable of capturing abrupt nonlinear voltage fluctuations during dynamic load switching. For long-term degradation prediction, TCN–LSTM is selected because dilated temporal convolutions efficiently capture slow degradation evolution while maintaining lower computational complexity. This pipeline-based integration ensures consistent feature representation across short-term and long-term prediction tasks.

Fig. 1 illustrates the voltage prediction framework of PEMFC using a combination of different approaches. In simple terms, the proposed system shows an integrated framework for fuel cell health monitoring by blending physically informed system, signal processing and artificial intelligence. The fuel cell monitoring system continuously senses measurable signals such as voltage, current, temperature and pressure.

These signals tell us how the system is operating in real time. The framework then analyses these signals in three different dimensions:

1) The model-based approach compares the real behaviour of the fuel cell with a mathematical or physical model. This captures system-level dynamics and electrochemical degradation mechanisms in real time.
2) The filter-based approach focuses on cleaning noisy signals and estimating hidden internal states using techniques like Kalman filter,

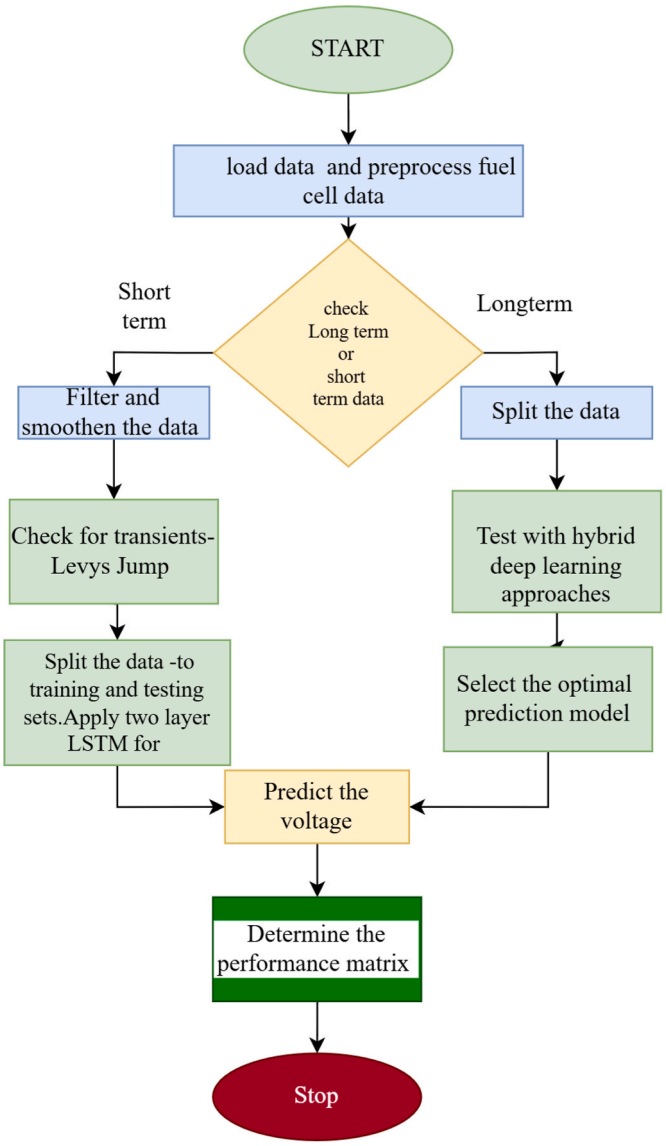


Fig. 2. Flow chart of the proposed methodology of PEMFC systems under consideration.

Table 1
Model Configuration and training setting of the Proposed TCN-LSTM Model for long term PEM Fuel Cell Voltage Prediction.

Parameters	Configuration	Selected Value
Data Pre-processing for long term prediction	Standardization	Standard scaler
Sequence Length	Temporal Window	30 time steps
TCN Layer 1	Dilation Factor	1
TCN Layer 2	Dilation Factor	2
TCN Channels	Hidden Units	64
TCN Kernel size	Convolution Filter Length	3
LSTM Layer	Hidden Units	64
Loss Function	Regression Loss	Mean Squared Error (MSE)
Regularization	L2 Weight Decay	1e-4
Optimizer	Adam	lr = 0.001
Adam Beta1	Momentum Parameter	0.9
Adam Beta2	Variance Parameter	0.999
Batch Size	Mini-batch	64
Training Epochs	Iterations	50

Table 2
Grid search Optimization values of the parameters used in the architecture for short term prediction in dynamic environment.

α	λ	δ	Validation RMSE
1.2	0.005	0.5	0.2638
1.5	0.01	1.0	0.2780
1.8	0.001	1.0	0.2904

Particle filters, Savitzky Golay etc. This helps track gradual performance changes more accurately while smoothing the signals sensed.

3) The data-driven approach uses deep learning to detect patterns in large amounts of historical and operational data. It learns complex relationships that are difficult to capture with physical models alone.

IEEE 2014 Dataset of PEM fuel cell embedded data set of nearly 1020-h duration is reported in Ref. [29]. The dataset is utilized for long term voltage prediction study and data csv file for short duration test voltage prediction is performed under controlled lab testing procedure at Electric Vehicle(EV) Lab setup. Output prognostics is to be done using the fused information that enables the system operator to perform Fault detection-detection of anomalies or abnormal behaviour in real time based on voltage prediction in two approaches enhancing reliability of predictions.

2.2. Multi-dimensional fusion strategy of the proposed PEMFC framework

The above mentioned three dimensions are united into one framework using equation (1)

$$F_{fusion} = \alpha F_{phys} + \beta F_{sig} + \gamma F_{dl} \quad (1)$$

Here F_{phys} stands for thermos fluid features (such as current, temperature); F_{sig} refers to signal-processed features (like Savitzky-Golay filtered voltage or rolling average); F_{dl} is a collection of deep features derived using proposed TCN–LSTM network. Here α, β and γ refer to fusion weights. In the proposed framework, the fusion procedure includes a cascade design with subsequent application of attention-inspired fusion weighting. Firstly, a concatenation of physically-informed and signal-processed features is applied:

$$X_{input} = [F_{phys}, F_{sig}] \quad (2)$$

Secondly, the fused input is passed through the deep-learning neural network to extract temporal features:

$$H_{short} = \text{two layer LSTM}(X_{input}) \quad (3)$$

$$H_{long} = \text{TCN} - \text{LSTM}(X_{input}) \quad (4)$$

Additionally, Lévy-distribution inspired residual enhancement is added to the loss function:

$$L = \text{Huber}(e) + \lambda |e|^{\alpha} \quad (5)$$

where $\alpha = 1.5$.

The proposed framework does not employ an independent secondary prediction network after feature fusion. Instead, the hybrid fusion process is integrated directly into the input representation stage of the deep learning architecture. The physically-informed features and signal-processed features are concatenated and supplied as input sequences to the temporal prediction networks. Therefore, the TCN–LSTM and two-layer LSTM architectures operate as the final end-to-end voltage predictors rather than standalone feature extractors. The fusion equation conceptually represents the multi-domain feature integration strategy used to construct the input feature space, while the final voltage prediction is generated directly by the deep learning models through their output regression layers. It increases the sensitivity of the network to heavy-tailed errors, making it more robust against load changes.

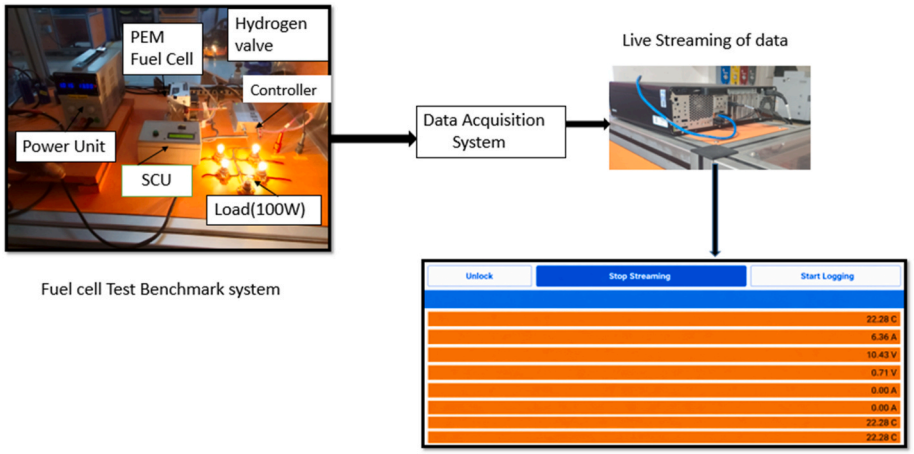


Fig. 3. Performance evaluation of 100W PEM Fuel cell in EV Lab.

Table 3
Fuel cell technical specification in laboratory test benchmark system.

Parameter	Value	Parameter	Value
Type of fuel cell	PEM	H_2 Pressure	0.45 – 0.55 bar
Number of cells & Rated Power	20 & 100 W	Hydrogen purity	≥99.995% dry H_2
Performance	12 V @ 8.3 A	Humidification	Self-humidified
Open Voltage	19.1 V	Stack weight (with fan & casing)	1460 g (±30 g)
Working Voltage	10.0 V – 16.0 V	Controller weight	400 g (±30 g)
PEM Stability after 12 h	12.8 V	External Dimension	11.8 × 10.4 × 9.4 cm
H_2 Supply valve voltage	12 V	Flow rate at max output*	1.3 L/min
Purging valve voltage	12 V	Start-up time	≤30 s
Anode reaction	$H_2 \rightarrow 2H^+ + 2e^-$	Efficiency of stack	40% @ 12 V
Cathode reaction	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	Low voltage shut down	10 V
Overall reaction	$H_2 + ½O_2 \rightarrow H_2O$	Over current shut down	12 A
External temperature	5 to 30°C	Over temperature shut down	65°C
Max. stack temperature	65°C	External power supply**	13 V (±1V), 5A
PEM Effective Area of Stack	26 cm^2		

Finally, attention coefficients are obtained implicitly by means of gradient backpropagation during training.

2.2.1. Data pre-processing and time-series generation of long term ageing data of PEMFC

In order to make the results of long-term voltage prediction using IEEE 2014 datasets fully reproducible, the following steps are taken in the pre-processing phase.

The missing values are identified by NaN detection. If there are any, then linear interpolation is used for their replacement:

$$V(t) = \frac{(V(t-1) + V(t+1))}{2} \tag{6}$$

Then, a Median Absolute Deviation (MAD) test is carried out for detecting possible outliers. The z-score in this test

$$Z_{robust} = 0.6745 \times \frac{(x - \text{median}(x))}{MAD} \tag{7}$$

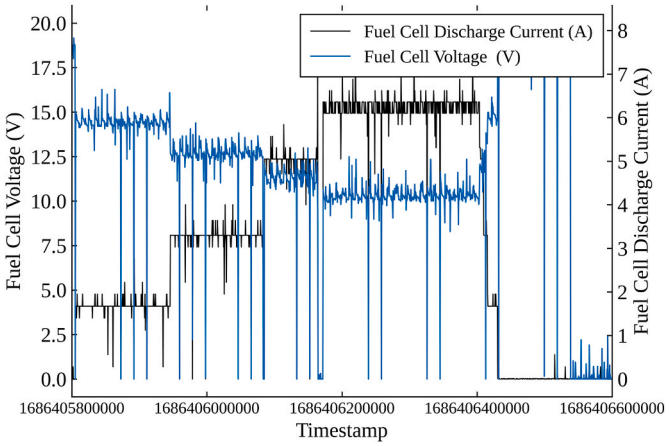


Fig. 4. Time series plot of Fuel cell voltage and current subjected to dynamic load.

$$\text{Where, } MAD = \text{median}\left(|x - \text{median}(x)|\right) \tag{8}$$

Data entries with $|Z_{robust}| > 3.5$ are regarded as outliers, which should be smoothed by using Savitzky-Golay filter or replaced by local interpolation.

Z-score standardization is employed for feature normalization to ensure stable convergence during deep learning training

$$x_{std} = \frac{x - \mu}{\sigma} \tag{9}$$

where μ and σ denote the mean and standard deviation of the training data, respectively. Standardization improves numerical stability and prevents feature dominance under varying operating conditions.

Temporal Modelling employs input/output sequences obtained through the sliding window method.

$$X_i = [x_i, X_{i+1}, \ldots, x_{i+L-1}], \tag{10}$$

$$y_i = x_{i+L} \tag{11}$$

where L = 30 is the sequence length and step size is 1. The long-term voltage prediction task is formulated as a one-step-ahead forecasting problem, where the voltage at time step (t+1) is predicted using the preceding 30 measurements.

Thereafter, the dataset is divided into training and testing parts according to an 80/20 split rule based on the chronological order. No random shuffling is involved in this procedure.

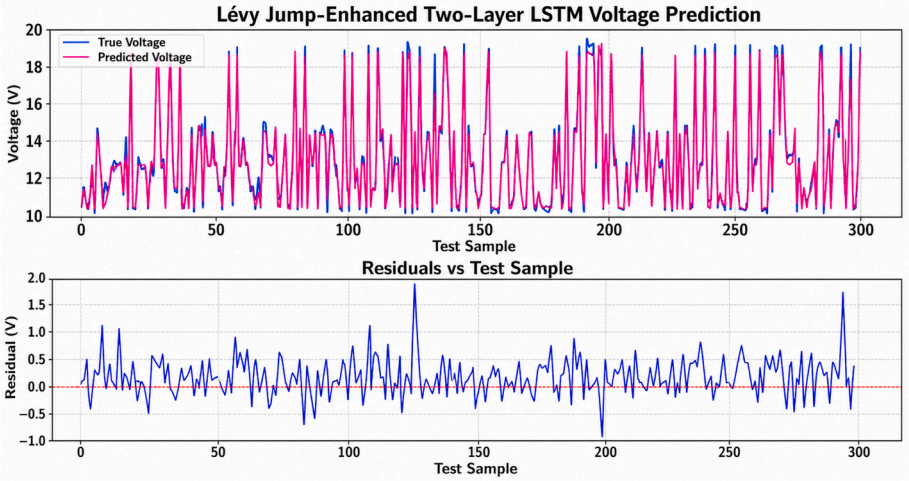


Fig. 5. a) Optimized 2 layer LSTM Voltage Prediction and b) residual error plot over time of laboratory test benchmark system.

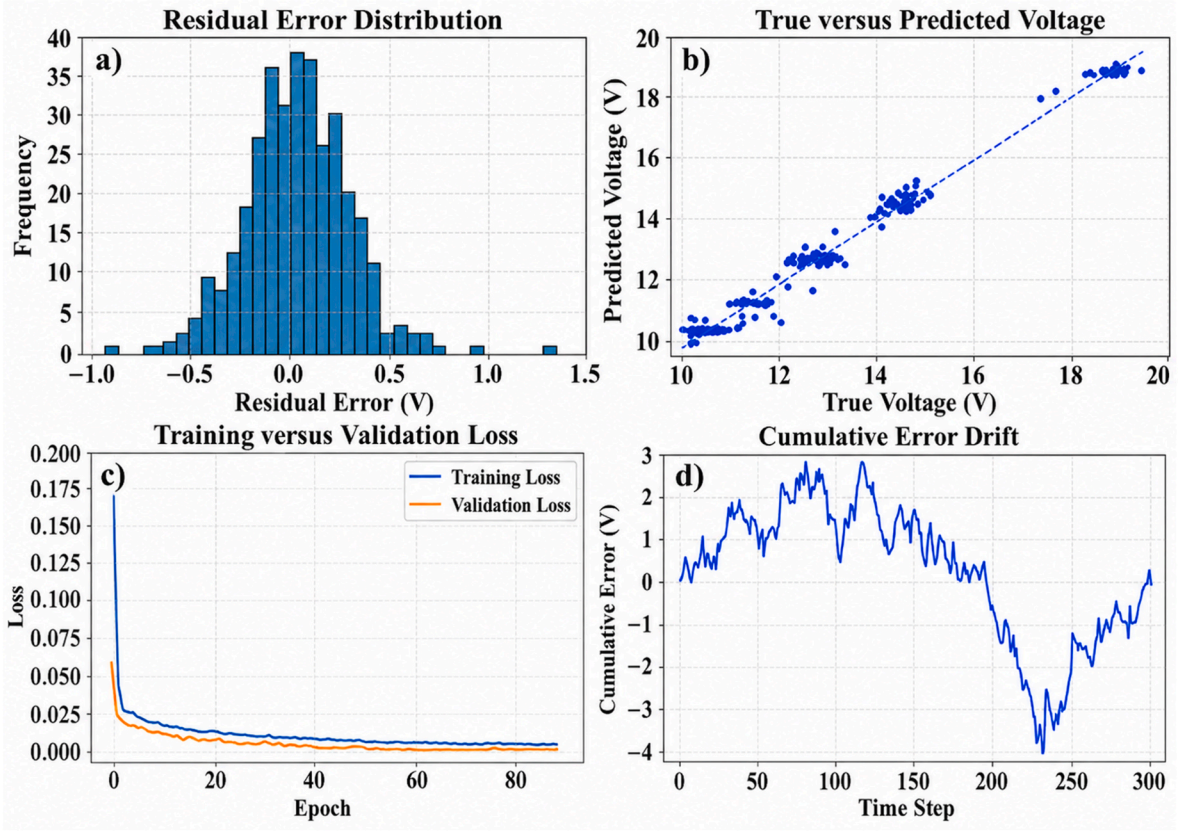


Fig. 6. 100W PEM Fuel cell laboratory test data based a) Residual error distribution with two-layer LSTM for voltage prediction b) True versus predicted voltage plot c) training and validation loss d) cumulative error drift waveform.

The proposed two-layer LSTM architecture is mathematically defined as:

$$h_1 = \mathrm{LSTM1}\left(X_{input}\right) \tag{12}$$

$$h_2 = \mathrm{LSTM2}(h_1) \tag{13}$$

$$\widehat{V} = W \cdot h_2 + b \tag{14}$$

where the first LSTM layer captures short-term transient voltage fluctuations, while the second layer models long-term temporal dependencies and recovery dynamics. This hierarchical structure improves the representation of both fast and slow electrochemical processes.

Fig. 2 illustrates the flow chart of the proposed methodology of voltage prediction using long term accelerated embedded ageing data and experimental short term data. Sampling rate for short-term data ($\geq$100 Hz) and Sampling rate for long-term data (1 Hz). The IEEE 2014 PEMFC Dataset is widely utilized as a standard for degradation modelling, providing extensive operational data at various timescales. The availability of short-term data through experimentation, as well as long-term benchmark data, allows for assessments to be conducted under varied operational regimes.

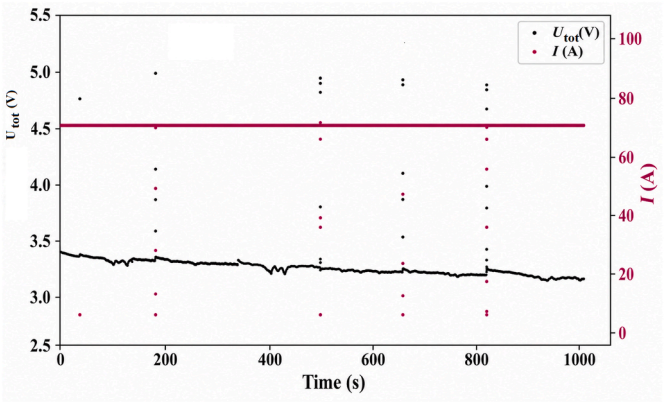


Fig. 7. Accelerated ageing data set Voltage and current waveform for 1020.54 h from IEEE 2014 challenge embedded dataset.

The voltage dynamics during dynamic loading are described in equation (15),where

$$V = E - V_{act} - V_{ohmic} - V_{conc} \tag{15}$$

V is Actual output voltage of the fuel cell, Reversible (ideal) voltage is denoted by E, V_{act} is activation loss of PEMFC that occurs due to electrochemical reaction kinetics (slow electrode reactions),V_{ohmic} is ohmic voltage loss and V_{conc} is mass transport loss. It dominates at low current densities as in Ref. [10].

2.2.2. Physical feature interpretation and mapping

The output voltage of a PEM fuel cell is affected by activation, ohmic and concentration polarizations. In the suggested approach, those physical phenomena are implicitly incorporated into the process of feature extraction via physically meaningful features. The voltage changes resulting from activation polarization (V_{act}), which is linked to kinetic phenomena, are captured using voltage difference and jump Magnitude features calculated according to the Lévy-based jump detection method, which accounts for the non-Gaussian nature of transients. Since ohmic polarization (V_{ohmic}) is caused by resistances in the membrane and electrodes, it is dependent on operating current and temperature. Therefore, the features representing ohmic behaviour include Current and Temperature. Finally, the effect of concentration polarization (V_{conc}), which is related to the phenomena of mass transfer and diffusion, is observed via slow voltage recovery and saturation. These aspects are represented using voltage rolling and regime-based features. Therefore, even though there is no direct enforcement of physical equations in the model training, the implicit embedding of physical knowledge allows maintaining physical meaning while utilizing data-based analysis.

$$V(t) = V_{ss} + (V_0 - V_{ss}) \cdot e^{-\frac{t}{\tau}} \tag{16}$$

V(t) = instantaneous voltage, V_{ss} is the steady state voltage of PEM fuel cell [13]. In this work, these physical equations are used to interpret voltage behaviour and validate model predictions, rather than being explicitly embedded into the model architecture or training process.

$$\tau = RC, \text{ the time constant} \tag{17}$$

Savitzky-Golay filtering is used to remove noise and retain dynamic degradation features for modeling. The Savitzky-Golay smoothing filter is a finite impulse response (FIR) smoothing filter and is also known as polynomial smoothing or least-squares smoothing filter without loss of information that can be implemented in FPGA [30].

Because of the load switching after 2 min in load steps of 20W, the short jump is identified by Levy's jump approach.

The observed jump information is incorporated into the learning model by enhancing the features of the system. The inputs for the model are formulated as in equation 18

$$X_{input}(t) = \left[T(t), I(t), V_{smooth}(t), V_{rolling}(t), J_{mag}(t), J_{event}(t)\right] \tag{18}$$

Here, $J_{mag}(t)$ refers to the magnitude of the jumps, whereas $J_{event}(t)$ indicates the binary value of jump events.

In order to have reproducibility, the Lévy jumps are identified according to a Median Absolute Deviation (MAD) based method. The robust Z-statistic is given by:

$$Z_{robust} = 0.6745 \times \frac{V(t) - \text{median}(V)}{M} \tag{19}$$

$$MAD = \text{median}(|V(t) - \text{median}(V)|). \tag{20}$$

If the inequality below holds:

$$|Z_{robust}(t)| > \theta \tag{21}$$

where the value of θ is 3.5, a jump occurrence is reported. This particular choice for the threshold is commonly used in statistical analysis and empirical studies reveal that this threshold reliably identifies voltage jumps during sharp changes in loads and does not report false jumps due to usual noise levels.

The magnitude of each jump (J_{mag} (t)) is quantified in equation (19).

$$J_{mag}(t) = |V(t) - V(t-1)| \tag{22}$$

while the binary jump indicator is calculated as given in (20)

$$J_{event}(t) \in \{0,1\} J_{event(t)} \in \{0,1\} \tag{23}$$

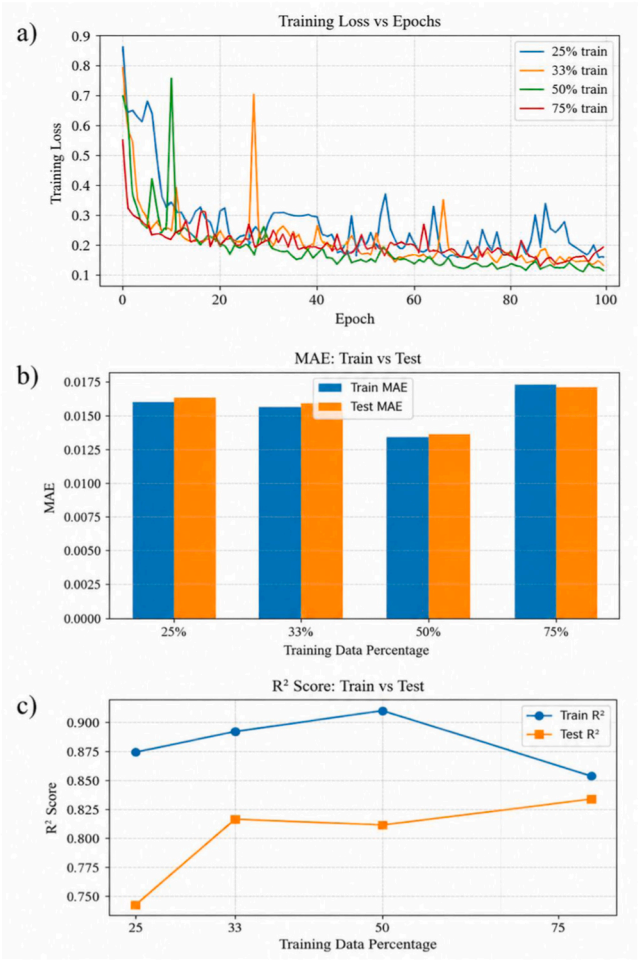
The rationale for including Lévy jump modelling is to capture the sudden and non-Gaussian voltage changes that happen when the load changes in the PEM fuel cells test benchmark system. Traditional LSTM models assume that time changes smoothly, which makes it hard for them to pick up sudden voltage spikes. The Lévy jump mechanism makes it possible for the model to take into account heavy-tailed transient behaviour, which makes it better at showing sudden electrochemical responses when the load is switched.

This work proposes a unified multi-timescale framework that combines experimental and benchmark datasets, unlike traditional methods that only look at short-term transients or long-term degradation separately. Additionally, combining Lévy jump-based transient modelling with LSTM makes it easier to capture non-Gaussian voltage dynamics when the load changes.

The proposed hybrid loss function is applied only to the short-term Lévy-enhanced LSTM model, whereas the long-term TCN-LSTM model is trained using the conventional MSE loss function shown in Table 1.

Table 1) denotes the parameters chosen for the selected model for long term degradation set. In order to guarantee stable convergence and efficient modelling of PEM fuel cell voltage dynamics under dynamic operating conditions, the chosen hyper parameters were established through a methodical empirical evaluation. Standardization reduces feature dominance and increases training robustness because PEM fuel cell voltage signals show nonlinear fluctuations under varying load and temperature conditions. To balance degradation memory and transient tracking, a sequence length of 30 time steps is chosen. Although lengthy windows increased computational complexity without significantly improving performance, shorter windows were insufficient to capture long-term drift behaviour. Both short-term load switching events and trends of gradual voltage degradation are well captured by the temporal window that is selected. The TCN architecture employs a kernel size of 3 with dilation factors of 1 and 2.

Multi-scale temporal feature extraction is made possible through the use of dilated convolution with dilation factors of 1 and 2. Rapid voltage dips and instantaneous temporal dependencies during abrupt load changes are captured by the first dilation level. The model captures delayed electrochemical recovery behaviour after transient disturbances



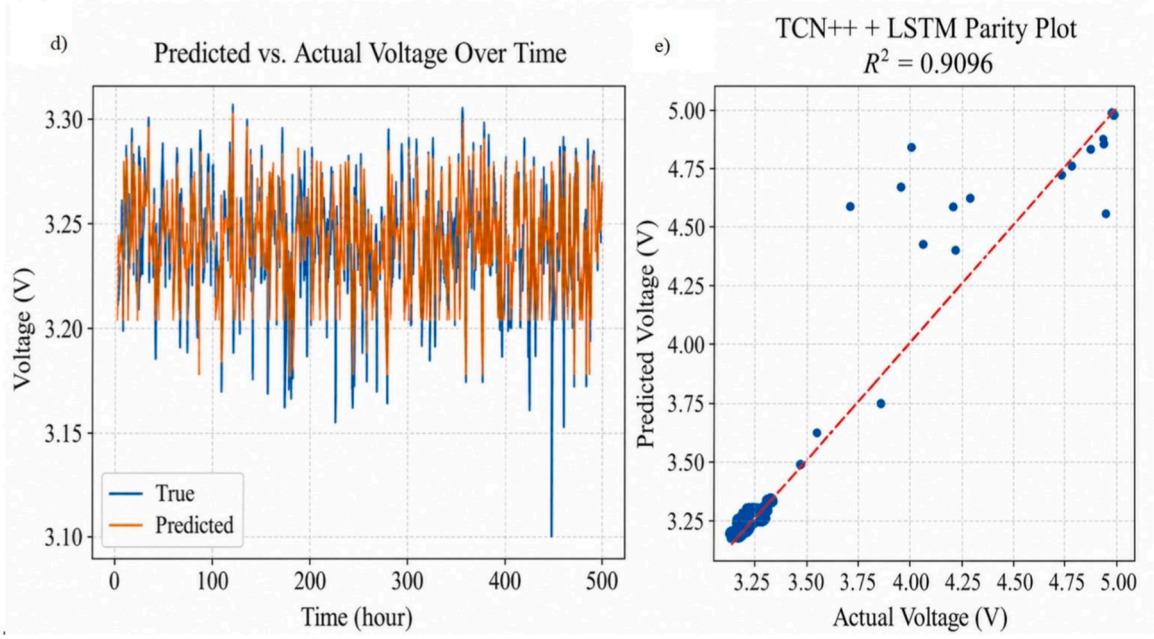


Fig. 8. a) b) c) Performance evaluation of proposed TCN- LSTM hybrid approach for various training and testing size. d) e) True and predicted Stack voltage during the ageing test by TCN- LSTM algorithm on embedded long term data.

Table 4
Comparison of Voltage prediction performance metrics in various hybrid deep learning approaches on ageing dataset.

Hybrid deep learning Method	R^2	RMSE	MAPE
AE-DNN	0.9007	0.0337	0.72
TCN-LSTM	0.9096	0.0220	0.0123
Q LSTM	0.869	0.553	0.8386
Transformer-LSTM	0.9109	0.0210	0.0137

in the second dilation level, which increases the receptive field without adding more parameters. In order to avoid over-parameterization and provide adequate representational capacity for nonlinear transient pattern learning, a channel size of 64 filters is chosen. In a similar approach, it is determined that 64 hidden units in the LSTM layer is sufficient to capture long-term voltage prediction. Because it penalizes large voltage deviations more severely and ensures accurate magnitude prediction during transient peaks and dips, Mean Squared Error (MSE) is chosen as the regression loss function.

L2 weight decay (1×10^{-4}) incorporated reduces overfitting caused by nonlinear temporal dependencies. This regularization strategy improves model generalization across varying operating conditions and load profiles. The adoption of Adam optimizer with a learning rate of 0.001 ensures stable convergence under highly nonlinear electrochemical signal behaviour. The momentum parameter ($\beta_1 = 0.9$) controls gradient smoothing and prevents oscillatory updates, while the second-moment parameter ($\beta_2 = 0.999$) stabilizes variance estimation and promotes steady convergence. A mini-batch size of 64 is selected to balance computational efficiency and training stability. Training for 50 epochs was found sufficient for empirical convergence while avoiding overfitting. Overall, the hyper parameter configuration is not arbitrarily selected but specifically tuned to address the multi-scale, nonlinear and transient characteristics inherent in PEM fuel cell voltage dynamics.

Even though hybrid TCN-LSTM architectures have been reported in time-series prediction literature, their systematic evaluation for PEM fuel cell voltage prediction under dynamic operating conditions remains limited. This study provides a structured benchmarking analysis of multiple deep learning architectures and demonstrates the effectiveness of an optimized TCN-LSTM configuration for capturing both transient voltage fluctuations and long-term degradation behavior.AE–DNN is an effective approach for long-term degradation trend learning but less sensitive to abrupt transitions. Transformer–LSTM Hybrid: Self-attention captures abrupt transitions while LSTM stabilizes sequential memory.

For the short-term Lévy jump-enhanced two-layer LSTM model, a hybrid loss function is employed during training to improve transient voltage spike tracking under dynamic load variations. The loss combines the robustness of the Huber loss with a fractional-order residual amplification term inspired by Lévy heavy-tailed behaviour. In contrast, the long-term TCN-LSTM model is trained using the conventional Mean Squared Error (MSE) loss function reported in Table 1. The hybrid loss function is formulated as:

$$L = L_{Huber} + \lambda |e|^{\alpha} \tag{24}$$

where $e = y_{true} - y_{pred}$. The Huber loss component is defined as:

$$L_{Huber} = 0.5e^2 \quad if\, |e| \le \delta$$

and

$$L_{Huber} = \delta\,(|e| - 0.5\delta)\ \text{if}\ |e| > \delta \tag{25}$$

Although the grid-search optimization given in Table 2 identifies $\alpha = 1.2$, $\lambda = 0.005$ and $\delta = 0.5$ as the minimum-validation-RMSE configuration, the final reported model employed $\alpha = 1.5$, $\lambda = 0.01$ and $\delta = 1.0$ because these parameters demonstrated more stable convergence behaviour and improved robustness during highly nonlinear transient voltage fluctuations. Delta(δ) is Huber threshold parameter is set to 1.0. MAD based robust Z score analysis is selected as $\theta = 3.5$. These parameters are determined through repeated validation experiments to achieve stable convergence and accurate transient voltage tracking under dynamic load-switching conditions. The proposed hybrid loss function is deployed as the optimization objective during training of the short-term Lévy-enhanced LSTM model. For model evaluation and comparison, standard performance metrics including MSE, RMSE, MAE and R^2 are reported. Therefore, the hybrid loss influences parameter optimization during training, whereas MSE serves only as a post-training evaluation metric.

The choice of $\alpha = 1.5$ is made owing to the capability of balancing robustness and sensitivity. Although $\alpha = 1$ implies MAE and $\alpha = 2$ implies MSE, taking $\alpha = 1.5$ would imply an intermediate characteristic that boosts sensitivity for high residual values without compromising computational efficiency. It is also consistent with the Lévy stable distribution, which is found to be an effective description of heavy-tailed stochastic processes similar to PEM fuel cells observations. The Levy jump mechanism combined with the enhanced two layer LSTM enables the models to capture heavy-tailed voltage spikes under dynamic load stress thenceforth improve the robustness. Two-Layer LSTM: Captures temporal dependencies and voltage recovery lag. Fuel cell voltage exhibits activation, ohmic and mass transport losses. Load steps introduce non-Gaussian transient spikes. Standard MSE under-penalizes these abrupt deviations. Pure L1 is too weak. Pure L2 is too sensitive to noise. The fractional exponent $\alpha = 1.5$ amplifies long-to-large residuals thereby preserves gradient stability, mimics Levy heavy-tailed stochastic processes and improves transient tracking without destabilizing training.

3. Results and discussion

The evaluation is performed across both short-term and long-term datasets to validate the effectiveness of the proposed multi-timescale framework under different temporal dynamics.

A. Short Term Voltage Prediction in Test Benchmark System

A laboratory-scale 100W PEM fuel cell stack tested under controlled step-load conditions is represented in Fig. 3. Voltage is measured continuously while dynamic load variations were applied up to 100W. The experiment duration was limited to 2 min due to for each load step. Step changes were introduced periodically to induce transient electrochemical responses, emulating real operational stress scenarios under controlled lab environment due to safety reasons.

3.1. Hardware specification and procedures

Table 3) represents the technical specification of hardware system

Table 5
Comparative Analysis of Deep Learning Architectures used in the long-term degradation study.

Rank	Model	Feature Extraction Method	Temporal Modeling	Long-Term Dependency	Noise Robustness	Computational Complexity
1	Transformer - LSTM	Self-Attention Encoder	LSTM	Very Strong	Very High	High
2	TCN - LSTM	Dilated Causal Convolutions	LSTM	Strong	High	medium
3	AE - DNN	Auto encoder (Nonlinear Compression)	Feedforward Only	Limited	Moderate	Low
4	Q-LSTM	Quantum-Inspired Gating	LSTM	Moderate	Low	High

for determining the dynamic load change. The experimental setup consists of a 100 W PEM fuel cell stack comprising 20 cells connected in series. Hydrogen gas is supplied from a compressed gas cylinder through a pressure regulator, which maintains the inlet pressure within the specified operating range of 0.45–0.55 bar. The purge valve removes the accumulated water and inert gases from the anode side and the hydrogen supply valve controls the fuel flow to the stack. To enable controlled short-circuit testing, a Short Circuit Unit (SCU) with an ON/OFF switch is integrated into the system for safety reasons. Since the stack is self-humidified, an external humidification system is not necessary. The blower ensures the oxidant supply to the cathode side and facilitates forced air cooling. Specific fuel cell load connectors and hydrogen input/output connectors, which are composed of silicon tubing to promote chemical compatibility and leak-tight performance, are used to interface the fuel cell stack with the external circuit. The stack is controlled by an on board controller that manages the operating conditions and interacts with the main switch and load switches for safe start-up, shutdown and load change. A Data Acquisition System (DAQ) is linked to the fuel cell terminals to facilitate continuous monitoring and recording of voltage, current and temperature readings at high sampling rates to capture both steady-state and transient responses. All acquired data is transmitted to a computer for real-time monitoring, storage and post-processing for model training and validation.

The experiments are conducted on a 100 W PEM fuel cell stack operated under dynamic load conditions. Hydrogen is supplied at controlled flow rate, while ambient air is provided at the cathode side. The stack is subjected to manual load variations using an electronic load to generate transient and steady-state operating regimes. Stack voltage (U_{tot}), current (I) and hydrogen inlet temperature (T_{inH2}) are recorded using a data acquisition system at uniform sampling intervals. The recorded signals were stored in CSV format for further processing. The dataset is pre-processed following the detailed pipeline described in Section 2.2.1.

3.2. Short term experimental voltage and current waveforms

Fig. 4 represents times series plot obtained from DAQ system. The transient response exhibits repeatable nonlinear properties that make it appropriate for simulating short-term voltage predictions. For short-term voltage prediction, a sequence length of 25 samples is adopted and the model is trained to perform one-step-ahead voltage forecasting. A sliding window with a step size of one sample is used to generate the short-term training sequences.

The short-term voltage prediction capability of the optimized two-layer Levy enhanced LSTM model using experimentally acquired and pre-processed PEM fuel cell data obtained under controlled laboratory conditions is illustrated in Fig. 5a. The improved transient tracking performance demonstrates the effectiveness of the Lévy jump-enhanced LSTM in capturing abrupt voltage fluctuations compared to conventional LSTM-based models. Fig. 5b denotes residual error plot of voltage prediction over specified time steps of experimentation.

Fig. 6 represents data driven performance analytics of 100W PEM Fuel cell laboratory test data. Fig. 6a Residual error distribution with two-layer LSTM for voltage prediction. The residual error histogram indicates that the prediction errors are roughly distributed around zero and there is no systematic bias in the prediction. Most of the prediction errors are confined to a small range and there are no large deviations in the prediction. Fig. 6b denotes True versus predicted voltage. Fig. 6c denotes both training and validation losses decrease rapidly in the initial epochs and then become stable and the model converges properly. The training and validation losses are very close to each other and there is no overfitting. Fig. 6d cumulative error drift waveform. The scatter plot shows that there is a strong linear relationship between the actual and predicted voltage values.

3.3. Long term voltage prediction using ageing embedded data

Fig. 7 represents the voltage and current variation in real time in embedded environment. The PEM fuel cell stack demonstrates a nonlinear dynamic behaviour. Activation and ohmic losses cause an instantaneous voltage drop upon abrupt current demand, which is followed by a prolonged recovery subjected to mass transport dynamics.

Fig. 8 represents the predictive performance of the proposed TCN–LSTM hybrid model for different training–testing splits (25%, 33%, 50% and 75%) using the IEEE 2014 PEM fuel cell challenge dataset. In Fig. 8a the training loss convergence behaviour is shown. For all data splits, the model demonstrates rapid convergence during the initial epochs, indicating effective learning of the nonlinear voltage dynamics. However, noticeable fluctuations are observed for the 25% and 33% training cases due to limited representation of transient operating conditions. The 75% training split exhibits smoother convergence and attains a lower final training loss compared to the other configurations. This indicates improved optimization performance when a larger portion of the dataset is used for training.

Fig. 8b compares the mean absolute error (MAE) for training and testing datasets. The 50% training configuration achieves the lowest test MAE, suggesting an appropriate balance between model complexity and data representation. The close agreement between training and testing MAE across all splits indicates good generalization capability with no significant overfitting. Fig. 8c shows the coefficient of determination (R^2) for both training and testing sets. The highest test R^2 is obtained with 80% training data, confirming superior predictive accuracy under this configuration. The model demonstrates consistent performance across different training–testing splits, indicating its robustness to variations in data distribution.

Fig. 8d and e illustrate the temporal evolution of the fuel cell stack voltage under long-term aging operation. A gradual downward trend in the voltage profile is observed, reflecting the natural degradation of electrochemical components such as the catalyst layer and membrane. Superimposed on this trend are several sharp voltage excursions, which correspond to operational transients such as start–stop cycles, load variations, purge events, or air/fuel supply disturbances. These abrupt variations highlight the highly non-stationary nature of real-world fuel cell operation and demonstrate the limitations of conventional degradation modelling approaches that rely solely on raw voltage time-series data.

As stated in Table 4) with R^2 of 0.9109 and RMSE of 0.0210, the Transformer - LSTM model performed the best in terms of prediction accuracy, as per the comparative study of performance. However, the difference in performance is very slight compared to the TCN - LSTM model (R2 = 0.9096, RMSE = 0.0220). Considering the fact that the computational complexity is much higher due to the self-attention mechanism in the Transformer model, the trade-off between accuracy and complexity in the TCN - LSTM model is preferable. Hence, the TCN - LSTM model is identified as an accurate and efficient model for long-term voltage prediction in a nonlinear operating condition. Although Transformer–LSTM achieves marginally higher prediction accuracy, TCN–LSTM is selected as the final architecture due to its lower computational complexity, stable convergence behaviour and suitability for real-time PEMFC applications.

In order to test the efficacy of the suggested hybrid loss function, a comparative study is carried out in relation to some widely-used loss functions such as MSE, MAE, and usual Huber losses using similar experimental settings. The findings show that although MSE gives excessive weight to noise, MAE gives less importance to large transient changes. In contrast, the suggested hybrid loss provides better performance when it comes to handling voltage changes.

Table 5 denotes suitability of four sophisticated deep learning architectures evaluates the long term voltage prediction under nonlinear operating conditions through a comparative analysis. Because of its potent self-attention encoder for feature extraction in conjunction with

LSTM-based temporal modelling, the Transformer - LSTM architecture received the best ranking. Strong long-term dependence capture and selective attention to significant temporal aspects are made possible by the self-attention mechanism. However, the Transformer - LSTM architecture exhibits significantly higher computational complexity due to the quadratic nature of the self-attention mechanism. Therefore, from a computational efficiency perspective, the TCN - LSTM architecture presents a more suitable alternative for resource-constrained or edge-oriented implementations.

4. Conclusion and future scope

The framework follows a physically-informed data-driven paradigm, where physical insights are integrated through feature representation and validation, ensuring a balance between model interpretability and computational efficiency for real-time applications. By transforming raw sensor measurements into meaningful degradation events, the proposed approach enables robust degradation tracking under highly dynamic operating conditions while significantly reducing computational burden.

The proposed work presents hybrid deep learning framework for short and long term horizon voltage degradation prognostics under dynamic load stress. The data acquired from the 100W PEMFC hardware test bench is validated for short term voltage prediction using Levy jump-enhanced two-layer LSTM. Further the embedded data is validated for long term voltage prediction using Auto Encoder-Deep Neural Network (AE-DNN), TCN -LSTM, QLSTM and Transformer-LSTM successfully. Although Transformer–LSTM achieved slightly higher prediction accuracy, TCN–LSTM demonstrated the best balance between accuracy, robustness and computational efficiency. Sequence-based and heavy-tail-enhanced architectures demonstrate superior performance under switching transient condition. The framework is scalable for future long-term aging and condition monitoring studies. Short-duration experiments capture early degradation signatures including transient voltage sag and recovery delay characteristics. Architecture suitability depends on degradation time scale and signal dynamics. The integration of short-term transient modelling and long-term degradation prediction demonstrates the capability of the proposed framework to operate across multiple temporal scales within a unified modelling strategy.

The proposed framework is well suited for real-time monitoring, digital twin applications and future uncertainty-aware prognostic extensions. For edge deployments in real-time PEM fuel cell voltage and state-of-health (SOH) prediction applications, the proposed TCN–LSTM framework shows strong potential for real-time PEMFC voltage and SOH prediction because it provides a favourable balance between prediction accuracy and computational efficiency for long term degradation data for condition monitoring of PEMFC systems. Even while the hybrid architecture is more computationally demanding than standalone recurrent models by nature, it is nonetheless feasibly deployable with the right architectural slimming techniques. Without sacrificing predictive effectiveness, computational economy is greatly increased by reducing model depth, restricting attention heads and limiting sequence duration. The new model explicitly links the process of electrochemical degradation to feature representation, while also proposing a mathematical-based fusion approach that utilizes physical features, signal processing, deep learning architecture and cascading architecture.

A limitation of this work is that physical knowledge is not directly embedded into the model training process and the use of a single benchmark dataset for long-term degradation analysis. Although the results demonstrate promising performance, further validation using datasets with diverse operating conditions is required to fully establish model generalizability. Future work will focus on incorporating physics-based constraints into the learning frameworks and validating the proposed framework on multiple datasets with varying distribution characteristics and real-world operational conditions to develop a digital twin framework.

CRediT authorship contribution statement

Asha Anu Kurian: Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Visualization, Writing – original draft. **Gnana Swathika O.V.:** Conceptualization, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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